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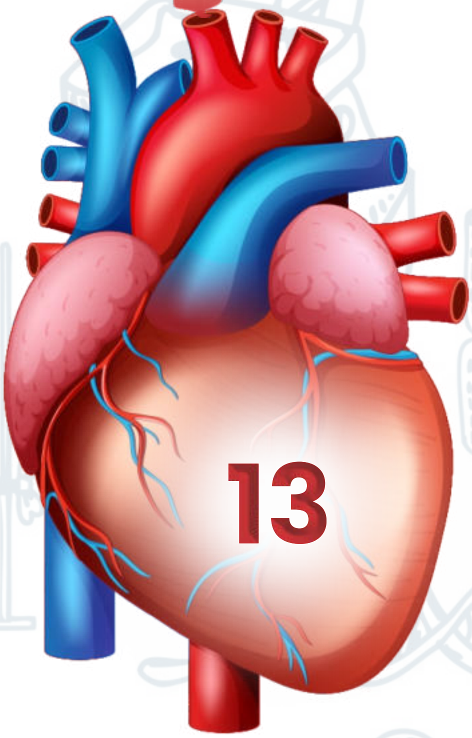
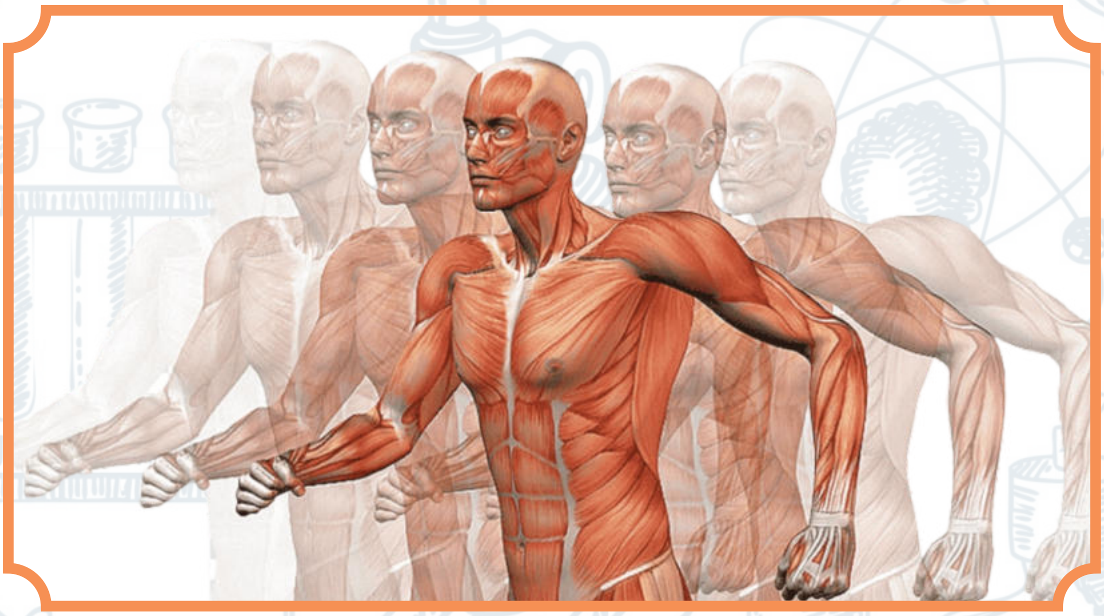


IMMUNITY



Immunity

Definition:

Immunity is the body's defense mechanism that protects it from pathogens (like bacteria and viruses), their toxins, and abnormal or foreign cells.

Functions of Immunity

1. Prevents Entry of Pathogens – Acts as a barrier against harmful microorganisms.
2. Destroys Invading Pathogens – Neutralizes or eliminates disease-causing agents that manage to enter the body.
3. Neutralizes Toxins – Detoxifies harmful substances released by microbes.
4. Removes Abnormal Cells – Identifies and destroys body's own damaged, mutated, or abnormal cells.
5. Rejects Foreign Cells – Fights against foreign cells, tissues, or organs (as in transplants).

Immune System

• A network of cells, tissues, and proteins that work together to protect the body from infection.
• Includes white blood cells (WBCs), antibodies, and complement proteins.
• Plays a critical role in:
• Fighting infections
• Preventing cancer
• Controlling allergies
• Managing autoimmune responses

Immunology

• The scientific study of the immune system and immunity.
• Helps understand:
• How the immune system functions.
• Why it may become overactive (allergies, autoimmunity).
• How it fails (immunodeficiency, cancers).
• The basis for vaccines and immune-based therapies.

Divisions of the Immune System

The immune system is functionally divided into two main types:

Innate Immune System (Non-Specific Immunity)

• **Nature:** Present from birth, provides immediate defense.
• **Non-specific:** Acts against any pathogen, not tailored to a specific invader.
• Two Lines of Defense:

First Line of Defense (Physical & Biochemical Barriers)

• Physical Barriers:
• Intact skin
• Mucous membranes
• Ciliated epithelium
• Biochemical Barriers:
• Saliva
• Mucus
• Tears
• Gastric juice (HCl)
• Spermine in semen

Second Line of Defense (Internal Non-specific Responses)

Cellular Defenses:
• Phagocytes (e.g., macrophages, neutrophils)
• Natural Killer (NK) cells

Other Responses:
• Inflammation
• Fever
• Antimicrobial proteins (e.g., interferons, complement proteins)

Adaptive Immune System (Specific Immunity)

• **Nature:** Develops over time, based on exposure to pathogens.
• **Specific:** Targets a particular pathogen with memory for future defense.

Third Line of Defense:

Specific Responses:
• **Cell-Mediated Immunity:**
• T lymphocytes (T cells) directly attack infected cells.
•• **Antibody-Mediated (Humoral) Immunity:**
• B lymphocytes (B cells) produce antibodies specific to antigens.

Summary Table: Three Lines of Defense

Line of Defense	Type	Key Components
1. First Line	Non-Specific	Skin, mucous membranes, saliva, tears, HCl, mucus
2. Second Line	Non-Specific	Phagocytes, NK cells, inflammation, fever, proteins
3. Third Line	Specific	B & T cells, antibodies (adaptive immunity)

First Line of Defense (Non-Specific Immunity)

Purpose: Prevents the **entry of all types of pathogens** into the body, regardless of their nature.

Types of Barriers Involved:

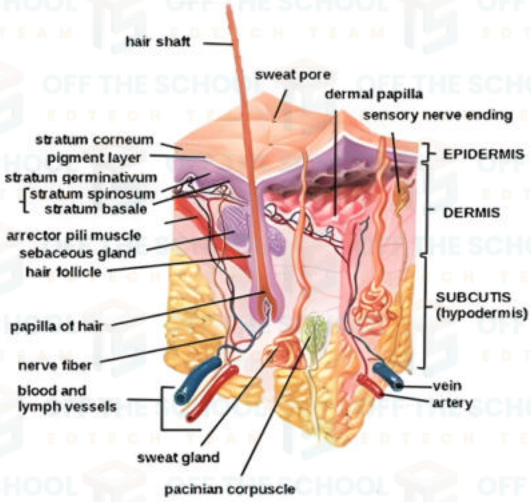
1. Physical Barriers:
• **Skin:** Acts as a physical wall, covering ~1.8 m² of the body. Protects against:
• Mechanical injuries
• Microorganisms
• Radiation, chemicals
• Temperature extremes

Also helps in:
• Body temperature regulation (via sweat and blood circulation)
• Fluid balance
• Synthesis of Vitamin D
• Sensory detection (touch, temperature, pain)

• **Mucous Membranes:** Line internal passages (e.g., respiratory, digestive tracts) and trap microbes.
• **Ciliated Epithelium:** Found in respiratory tract; helps in moving trapped pathogens out via mucus.

2. Biochemical Barriers:

• **Saliva:** Contains enzymes like lysozyme that break down bacterial cell walls.
• **Tears:** Contain antimicrobial enzymes.
• **Mucus:** Traps pathogens and particles.
• **Gastric Juice (HCl):** Destroys microbes ingested with food.
• **Spermine in semen:** Has antimicrobial properties.



1. Epidermis (Outer Layer)

• **Outermost layer of the skin;** composed of **closely packed dead cells.**
• Contains **keratin,** making it:
• Mechanically tough
• Resistant to bacterial enzymes and degradation
• **Lacks blood vessels:** depends on the dermis for nutrient/waste exchange.
• **Fatty acids** on skin surface make it **dry, salty, and acidic,** which:
• Inhibits microbial growth
• Acts as a chemical barrier
• **Dead skin cells are frequently shed**, removing attached microbes.
• Contains **immune cells** (e.g., **resident macrophages, T-cells**) for added protection.

2. Dermis (Middle Layer)

• Located **beneath the epidermis**.
• Contains:
• **Living cells, blood vessels, nerves**
• **Sweat glands:** secrete antimicrobial peptides
• **Sebaceous glands:** secrete **sebum** (oil), which:
• Lubricates the skin/hair
• Forms a protective **acidic film** on skin, deterring microbial growth

3. Hypodermis (Subcutaneous Layer)

Deepest skin layer, made up of:
• **Adipose (fat) tissue**
• **Connective tissue**

Functions:
• Connects skin to muscles/bones
• Acts as an insulating and cushioning layer

Second Line of Defense: Innate Immune Response

Pathogen Entry
• **Entry points:** Pathogenic microorganisms can enter the body through:
• **Broken skin** (due to cuts, burns, etc.)
• **Insect bites**
• **Inhaled air**
• When the **first line of defense** (skin, mucous membranes, etc.) is breached, the **second line of defense** is activated.

Components of the Second Line of Defense
• Comprised primarily of **White Blood Cells (WBCs)** and various **defensive proteins**.
• **Non-specific** in nature — responds broadly to invaders, not specific pathogens.

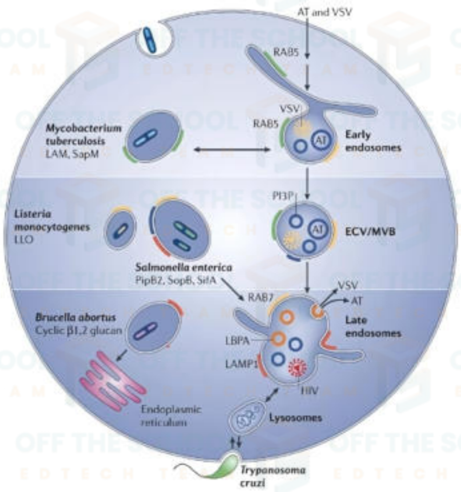
Major Responses of the Second Line of Defense

1. Phagocytosis
• Performed by neutrophils, monocytes, macrophages, and dendritic cells.
• Engulfs and digests pathogens or debris.

2. Cytotoxic Activity
• Carried out by **cytotoxic cells** such as:
• Natural Killer (NK) cells
• Cytotoxic T cells
• Destroy infected or cancerous cells.

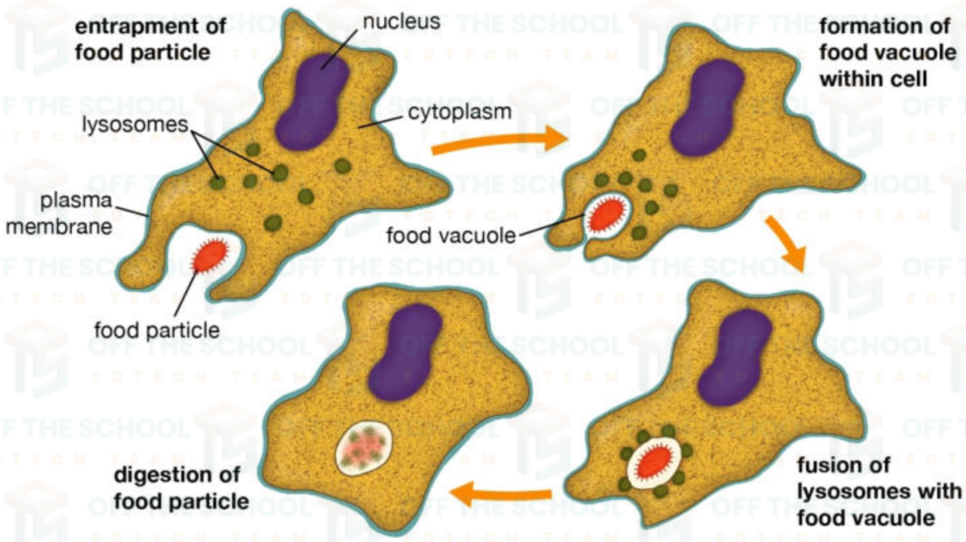
3. Inflammation
• Triggered by tissue damage or infection.
• Symptoms: redness, swelling, heat, pain.
• Purpose: isolate infection, attract immune cells, and promote healing.

4. Fever
• A systemic response to infection.
• Raised body temperature helps:
• Inhibit pathogen growth.
• Boost immune efficiency.



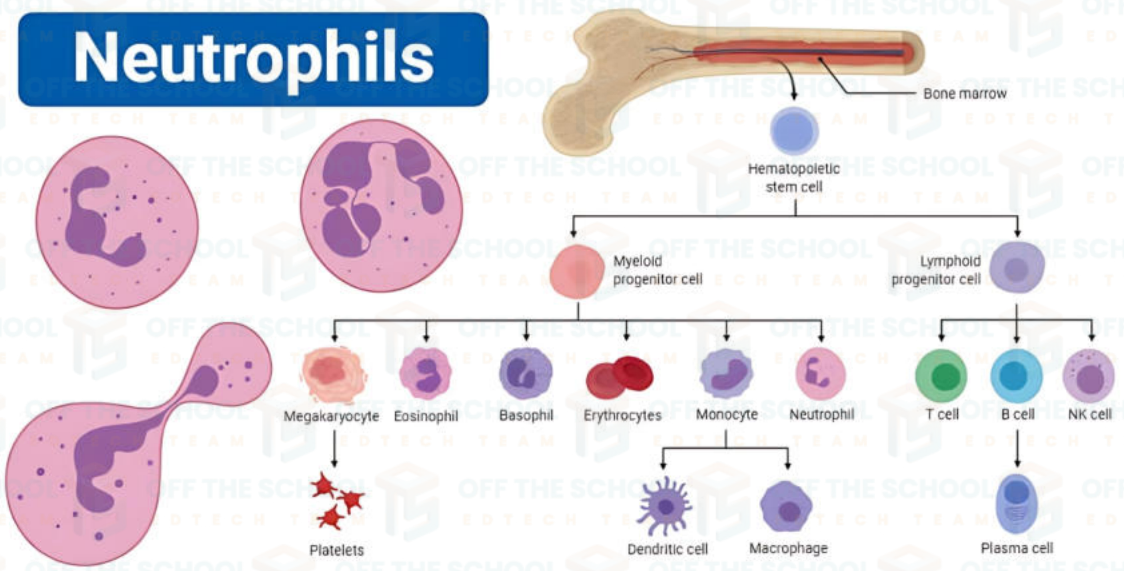
Phagocytosis

• **Definition:** Phagocytosis is the process by which certain white blood cells (WBCs) ingest and digest foreign particles, such as microorganisms and cellular debris.
• **Main phagocytic cells:** Neutrophils, monocytes (which differentiate into macrophages and dendritic cells).



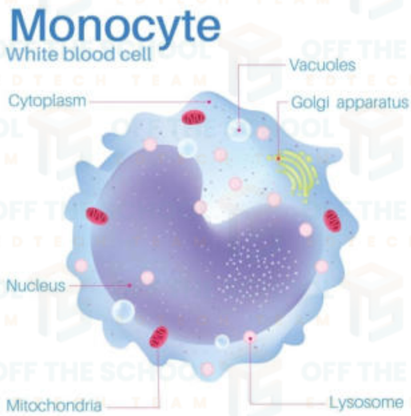
Neutrophils

• **Abundance:** Most abundant WBCs (50–70% of all leukocytes).
• **Structure:** Multi-lobed (3–5 lobes) non-spherical nucleus.
• **Lifespan:** Short-lived; migrate from blood into tissues upon maturation.
• **Function:** Specialized in bacterial phagocytosis.
• **Mechanism:** Possess surface receptors that recognize bacterial components (e.g., peptidoglycans, flagellin, lipopolysaccharides).
• **Role:** First line of defense in innate immunity.



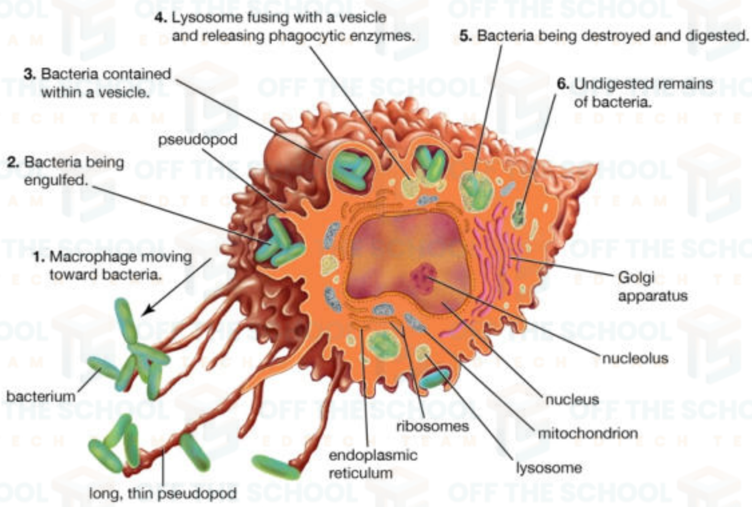
Monocytes

• **Abundance:** Constitute 2–8% of total WBCs.
• **Structure:** Bean-shaped nucleus, non-granulated cytoplasm.
• **Origin:** Derived from myeloid stem cells.
• **Lifespan:** 2–5 days in blood; longer in tissues after differentiation.
• **Function:** Precursors to macrophages and dendritic cells; involved in innate immunity.



Macrophages

• **Origin:** Differentiated monocytes that migrate into tissues.
• **Structure:** Large cells with indented, horseshoe-shaped nucleus.
• **Location:** Present in lymphoid and non-lymphoid tissues.
• **Functions:**
• Phagocytose foreign particles and dead cells.
• Present antigens to **T-cells** (antigen-presenting cells).
• Secrete **cytokines** and **nitric oxide** (kills pathogens).
• Adapt to specific tissue environments (e.g., alveolar macrophages in lungs, Kupffer cells in liver).
• **Structure:** Have long cytoplasmic projections (dendrites).
• **Function:**
• Phagocytose pathogens and present antigens to **T-cells.**
• Migrate to secondary lymphoid organs (tonsils, Peyer's patches, spleen) after activation.
• Unlike macrophages, they die after activating T-cells.



Natural Killer (NK) Cells

- **Type:** Lymphocytes (not phagocytic).
- **Function:**
- Kill **virally infected** and **cancerous** cells.
- Do not require prior activation by antigen-presenting cells.
- Part of **innate immunity**.

Natural Killer (NK) Cells and Their Role in Immunity

• Activation of NK Cells: NK cells are activated to cause lysis (destruction) of target cells. They are part of the body's innate immune system and play a critical role in identifying and eliminating abnormal or infected cells.

Mechanism of Action:

• NK cells act similarly to a "policeman," checking the surface of cells for identification, specifically looking for a molecule called **MHC-I (Major Histocompatibility Complex I)**.

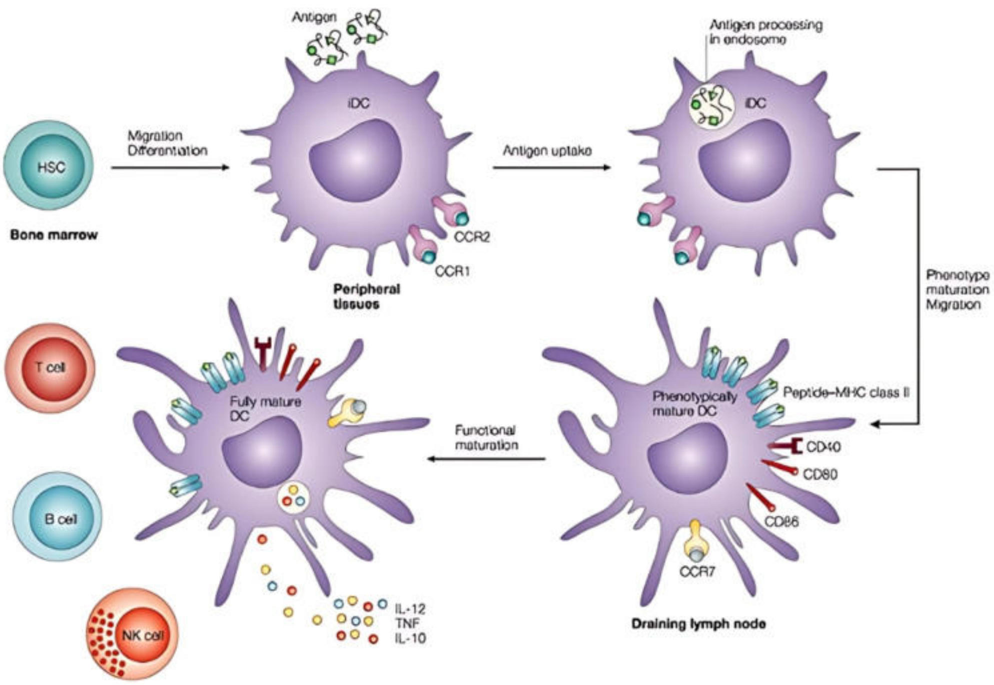
• **MHC-I** serves as an identification marker on normal cells. If a cell has normal MHC-I expression, the NK cell remains inactive.

• However, if there is any modification, absence, or abnormality in the MHC-I molecule, the NK cell becomes activated and targets the cell for destruction.

Destruction of Target Cells:

• Upon activation, NK cells destroy the target cell through the release of cytotoxic **granules**.

- These granules contain:
- **Perforin:** Forms pores in the plasma membrane of the target cell.
- **Granzymes:** Enter the target cell and trigger apoptosis (programmed cell death), damaging the target cell's internal structures, including its membrane.



Cytotoxic T Cells (Tc Cells)

Overview:

• Cytotoxic T Cells (Tc Cells) are part of the immune system and belong to the lymphocyte family of white blood cells (WBCs).
• As the name suggests, they are specialized in recognizing and attacking virally infected cells as well as tumor cells.

Origin and Development:

• **Production:** Cytotoxic T cells are produced in the bone marrow.
• **Maturation:** After being produced, they migrate to the thymus for further development. In the thymus, they mature and acquire specific surface markers, including the CD8 co-receptor, which helps them recognize infected or abnormal cells.

Maturation and Co-receptors:

• During maturation, Cytotoxic T cells develop two key components:
1. T Cell Receptor (TCR): The receptor that recognizes antigens on the surface of infected or abnormal cells.
2. CD8 co-receptor: This co-receptor helps the Cytotoxic T cells recognize and bind to MHC-I molecules on the surface of target cells.

Mechanism of Action:

• **Recognition:** Tc cells recognize virally infected cells or tumor cells through their TCR, which binds to fragments of proteins (antigens) presented by the MHC-I molecules on the surface of infected or abnormal cells.
• **Activation:** Once a Cytotoxic T cell recognizes its target cell, it becomes activated and can release cytotoxic granules.
1. These granules contain perforin, which forms pores in the target cell's plasma membrane, and granzymes, which enter the target cell and trigger apoptosis (programmed cell death).

Interaction with Helper T Cells:

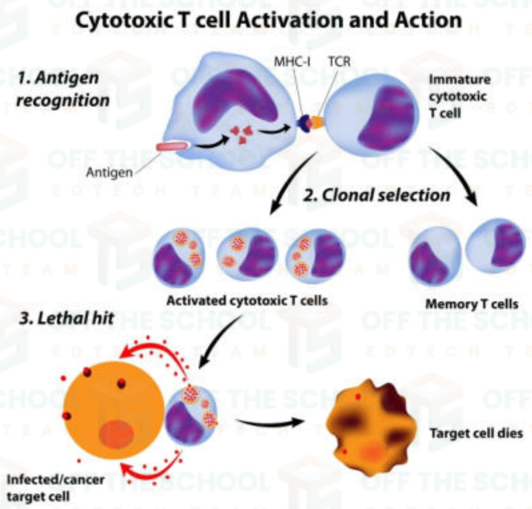
• Helper T Cells (Th cells) play a crucial role in activating Tc cells.
• The interaction between Tc cells and helper T cells is essential for the Tc cells to be fully activated and efficiently attack infected or abnormal cells.
• Helper T cells release cytokines (e.g., Interferon-gamma (IFN-γ)) that support Tc cell activity and enhance the immune response.

Actions of Cytotoxic T Cells:

• Lysis of Target Cells: Upon activation, Tc cells destroy the target cell (infected or tumor cell) through:
1. Perforin: Forms pores in the plasma membrane of the target cell.
2. Granzymes: Enter the target cell and trigger apoptosis, leading to the destruction of the infected cell.
• Unlike NK cells, which target cells indiscriminately, Tc cells are highly specific to the infected or abnormal cells they recognize via the MHC-I pathway.

Role of Cytokines:

• **Interferon-gamma (IFN-γ):** Released by activated Tc cells and helper T cells, this cytokine can help enhance the activity of other immune cells, such as macrophages and dendritic cells, in eliminating infections or tumor cells.



Protective Proteins

Complement Proteins:

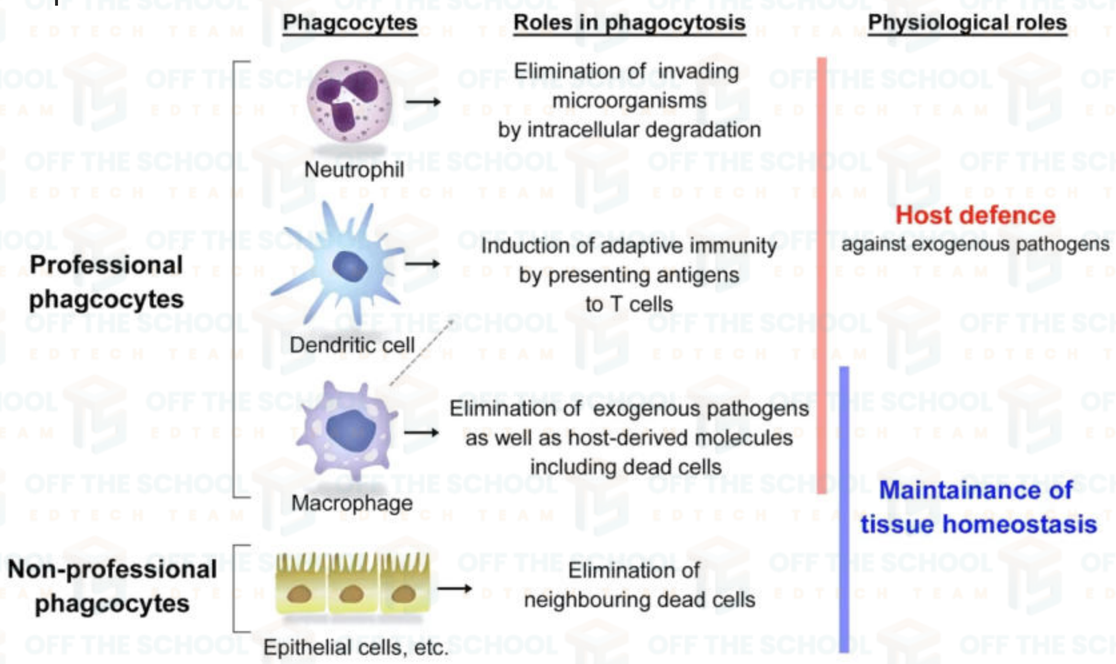
• **Definition:** Complement proteins are a group of proteins circulating in body fluids, which help the immune system combat infections, including bacterial infections.

• **Activation:** These proteins are activated through a series of reactions, typically involving foreign pathogens like bacteria.

Functions:

• **Damaging Target Cells:** Some complement proteins directly damage the plasma membranes of infected cells, leading to their destruction.

• **Attracting Immune Cells:** Other complement proteins help to recruit immune cells like mast cells, neutrophils, and macrophages to the site of infection, enhancing the immune response.



Inflammatory Response

Overview:

• Inflammation is a non-specific immune response typically caused by infection, tissue damage, irritants, or autoimmune disorders.

• It is the body's way of protecting and healing affected tissues, but it can also contribute to disease if misdirected.

Types of Inflammation:

Acute Inflammation:

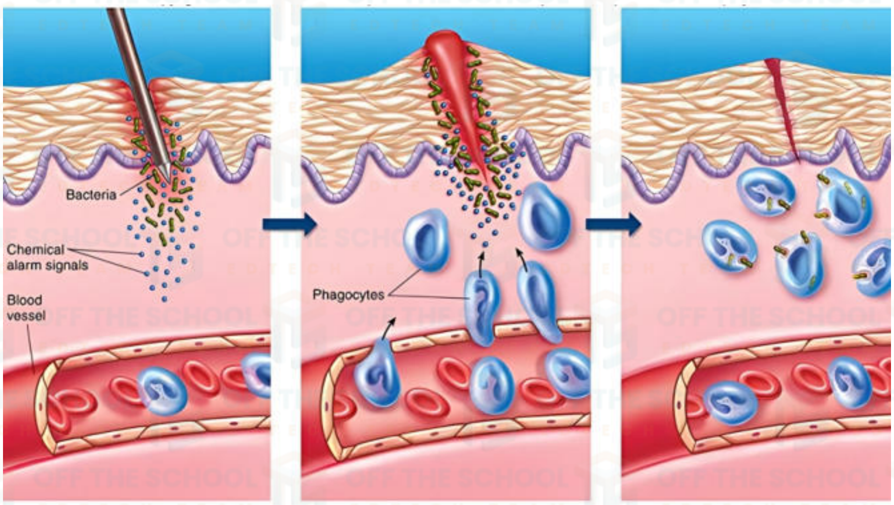
Characteristics: Short-term inflammation with visible symptoms such as redness, heat, pain, swelling, and loss of function of the affected organ.

Chronic Inflammation:

- **Characteristics:** Long-term inflammation, often due to chronic diseases like Cardiovascular Disorders (CVD) or allergies.

Inflammatory Process:

- **Tissue Damage:** When tissue is damaged (e.g., from an injury or infection), harmful bacteria can enter the body.
- **Release of Proteins:** The damaged tissue releases proteins (such as histamine and prostaglandins) that trigger inflammation. These proteins cause nearby blood vessels to dilate, increasing blood flow to the affected area.
- **Immune Response Activation:** Histamine and prostaglandins attract immune cells to the site of infection or injury, where they help eliminate bacteria through phagocytosis.
- **Pus Formation:** If tissue cells die during the inflammatory response, they may form a thick, paste-like substance called pus, which can have a foul odor. The pus is composed of dead cells, bacteria, and tissue debris.



Temperature Response or Fever (Pyrexia)

Definition of Fever:

- Fever is when the body's temperature rises above normal levels. It is not a disease, but a symptom that can indicate an underlying issue, such as an infection.
- It is a non-specific immune response triggered by various factors, including infections or external agents like medications and toxins.

Causes of Fever:

- **Endogenous Pyrogens:** These are substances produced by the body itself (e.g., cytokines secreted by immune cells).
- **Exogenous Pyrogens:** These are external agents, such as bacteria, viruses, or toxins, that can trigger fever when they invade the body.

Mechanism of Fever (How it Works):

- **Pyrogens:** When the body detects pathogens (e.g., bacteria or viruses), immune cells like macrophages release certain pyrogens (e.g., Interleukin-1 (IL-1), Interleukin-6 (IL-6), and Tumor Necrosis Factor (TNF)).

• These pyrogens travel through the bloodstream to the hypothalamus (the part of the brain that regulates body temperature).
• In the hypothalamus, pyrogens trigger the release of prostaglandins, which act as signaling molecules that increase thermogenesis (heat production) and decrease heat loss.
• This results in a rise in body temperature, leading to fever.

Benefits of Fever:

• Inhibits Pathogen Growth: Many fungi and bacteria cannot grow or reproduce at higher temperatures, so fever can help slow the progression of infections.
• Enhances Immune Response: The increased body temperature helps in the phagocytosis of pathogens by immune cells, improving the immune system's ability to fight infections.

Negative Effects of Fever:

• **Energy Loss:** A considerable amount of energy is expended as heat during fever.
• **Symptoms:** Fever can cause discomfort like fatigue, dehydration, body aches, and even seizures in severe cases.
• **Enzyme Denaturation:** Temperatures higher than 105°F (40.5°C) can denature enzymes and other proteins in the body, disrupting normal cellular functions.
• **Cell Damage:** High temperatures can cause damage to body cells and can potentially lead to organ failure or death in extreme cases.
• **Antipyretic Drugs:** Why Physicians Prescribe Them

Antipyretic Drugs:

• These are medications that reduce or control fever. They work by inhibiting the production of prostaglandins, which are responsible for the fever response.
• Prostaglandins are secreted in the hypothalamus and play a key role in raising body temperature. Antipyretics block the enzyme cyclooxygenase (COX), which is involved in the production of prostaglandins.
• Categories of Antipyretic Drugs:

Salicylates:

• **Example:** Aspirin.
• Aspirin works by inhibiting COX enzymes, reducing inflammation and fever.

Acetaminophen:

• **Example:** Paracetamol.
• It also reduces fever but works slightly differently than aspirin, primarily acting in the brain to reduce the set point temperature in the hypothalamus.
• Non-Steroidal Anti-Inflammatory Drugs (NSAIDs):
• **Example:** Ibuprofen.
NSAIDs reduce fever and inflammation by blocking COX enzymes, similar to aspirin.
• Reason for Prescribing Antipyretics:
• While fever is a natural immune response, it can become problematic if it is too high or lasts for an extended period. In such cases, antipyretics are prescribed to:
• Prevent damage to the body (such as denaturation of enzymes and organ damage).
• Alleviate symptoms such as fatigue, dehydration, and discomfort.
• Lower body temperature to a safer level, especially if the fever reaches dangerous levels (over 105°F).

Third Line of Defense – Specific (Adaptive) Immunity

Overview:

• Works alongside the second line of defense.
• Also called the Acquired or Adaptive Immune System.
• Highly complex, specific, and personalized (varies from person to person).
• It is the most powerful and precise mechanism for fighting infections.

Key Features:

• **Specific:** Targets particular pathogens.
• **Adaptive:** Learns and remembers pathogens.
• **Long-term protection:** Develops memory cells for faster future response.
• **Cellular players:** Mainly B-lymphocytes (B-cells) and T-lymphocytes (T-cells).

Recognition of Self vs Non-Self:

• Immune system must distinguish between self (body's own cells) and non-self (invaders).
• This recognition is made possible by antigens on the surface of foreign organisms.
• **Antigens:** Usually foreign proteins that trigger an immune response.

Role of B-lymphocytes:

• Recognize specific antigens.
• Produce antibodies – small, specific proteins that bind to the matching antigen.
• Antibody-antigen binding neutralizes or destroys the pathogen.

Memory and Reinfection:

• After first exposure, the immune system remembers the pathogen.
• Upon reinfection, it can quickly produce the same antibodies, offering immunity.

Major Histocompatibility Complex (MHC)

What is MHC?

• MHC (Major Histocompatibility Complex) is a group of genes that code for proteins found on cell surfaces.
• These proteins help the immune system recognize self vs non-self.
• In humans, MHC proteins are also known as Human Leukocyte Antigens (HLA).

Classes of MHC:

There are three classes, but the focus is mainly on MHC Class I and MHC Class II.

1. MHC Class I (MHC-I):

• **Location:** Present on all nucleated cells and platelets.
• **Structure:** Composed of:
• An alpha chain (α1, α2, α3 domains)
• A β2-microglobulin protein
• A binding groove between α1 and α2 for holding antigens

Function:

• Presents antigens from inside the cell (e.g., viral proteins).

Interacts with:

• Cytotoxic T cells ($CD8^+$ / Tc cells)
• Natural Killer (NK) cells via inhibitory receptors
• If MHC-I is altered, absent, or shows abnormal peptides, the immune system marks the cell as "non-self" and destroys it.

• Acts as a surveillance signal to immune cells.

2. MHC Class II (MHC-II):

• **Location:** Found only on antigen-presenting cells (APCs):
• Macrophages
• Dendritic cells
• B-lymphocytes

Structure:

• Made of two alpha chains and two beta chains (heterodimer)
• Contains a shallow, open-ended binding groove

Function:

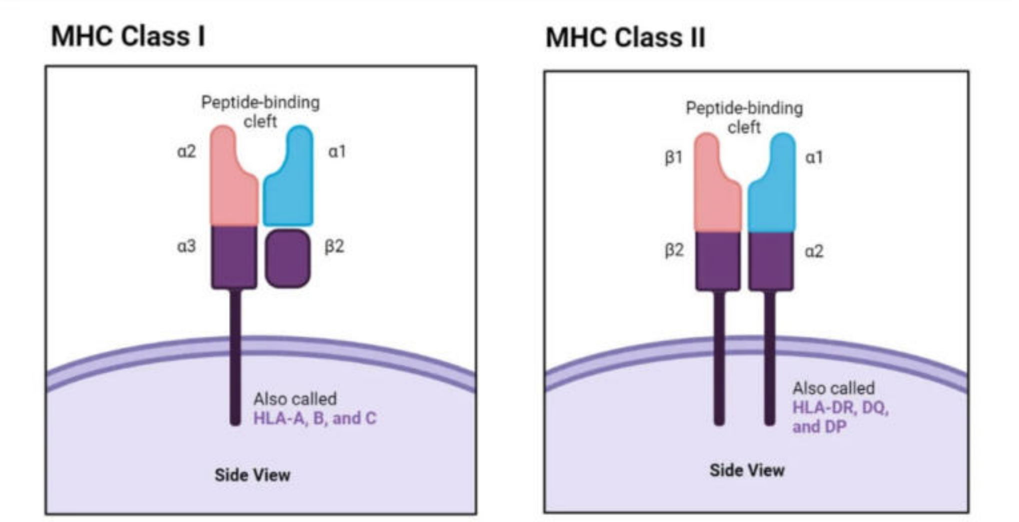
• Presents external antigens (e.g., bacterial fragments) to:
• Helper T cells ($CD4^+$)
• If a specific antigen is displayed on MHC-II:
• The naïve Helper T cell is activated
• Triggers activation of other immune cells and development of an adaptive immune response

Summary Table:

Feature	MHC Class I	MHC Class II
Found on	All nucleated cells, platelets	Antigen-presenting cells (APCs)
Interacts with	$CD8^+$ T cells, NK cells	$CD4^+$ T cells (Helper T cells)
Antigen Source	Intracellular (e.g., viruses)	Extracellular (e.g., bacteria)
Function	Marks self / non-self	Activates adaptive immunity

Types of Immunity

The human body uses two main types of immunity to protect against diseases:



1. Innate Immunity (Inborn Immunity):

Definition:

A non-specific defense mechanism present from birth. It provides immediate protection against all kinds of pathogens.

Key Characteristics:

• Present at birth (genetically inherited).
• Non-specific – acts the same way for all pathogens.
• Provides first line and second line of defense.
• Does not improve with repeated exposure.

Components:

• **Physical Barriers:**
• Skin
• Mucous membranes
• **Cellular Barriers:**
• Neutrophils
• Macrophages
• Mast cells
• Natural Killer (NK) cells
• **Chemical Barriers:**
• Cytokines
• Interferons
• Interleukins
• Enzymes in body fluids (e.g., saliva, tears)

Function:

• Rapidly responds to invaders.
• Prevents pathogens from entering or spreading in the body.
• Acts as a first alert system for the immune response.

2. Acquired Immunity (Adaptive Immunity):

• Mentioned but not elaborated in the text.
• Develops after exposure to pathogens.
• Specific to particular antigens.
• Forms immunological memory for future protection.
• Not present at birth – develops after exposure.
• Highly specific to a particular pathogen.
• Slower to respond initially (compared to innate immunity).
• Generates memory cells for quicker response during reinfection.

Components:

• **Lymphocytes:**
• B-lymphocytes (B cells): Produce antibodies that bind to antigens in body fluids.
• T-lymphocytes (T cells):
• Helper T cells ($CD4^+$): Activate other immune cells.
• Cytotoxic T cells ($CD8^+$): Kill infected cells.
• **Antibodies:** Proteins that specifically bind to antigens and help neutralize or destroy them.
• **Memory cells:** Long-lived B and T cells that respond faster upon future exposure.

Types of Acquired Immunity:

Active Immunity:

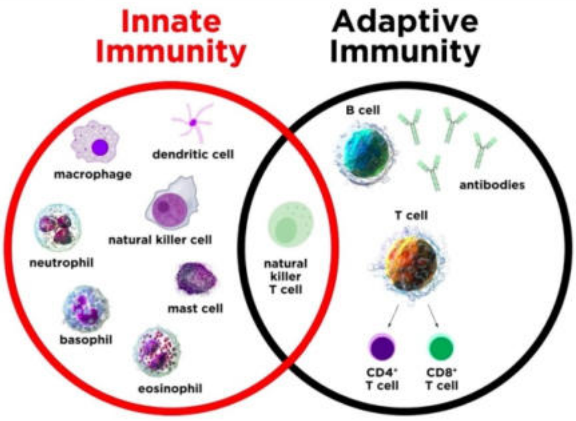
- **Natural:** Developed after recovering from an infection.
- **Artificial:** Gained through vaccination.

Passive Immunity:

- **Natural:** From mother to child (e.g., via placenta or breast milk).
- **Artificial:** Through injection of antibodies (e.g., antiserum).

Summary Table:

Feature	Innate Immunity	Acquired Immunity
Present at birth	Yes	No (develops over time)
Specificity	Nonspecific	Highly specific
Response time	Immediate	Slower (first time)
Memory formation	No	Yes
Involves	Physical, chemical, cellular barriers	B and T lymphocytes, antibodies
Type of protection	General, broad	Targeted, long-lasting



Types of Acquired Immunity:

Type	Natural	Artificial
Active	Infection (clinical or subclinical)	Vaccination (e.g., live attenuated, killed, or purified antigen vaccines)
Passive	Transfer from mother (e.g., via placenta or breast milk)	Injection of immune serum or immune cells from another person or organism

Active Immunity:

- Body produces its own antibodies.
- Takes time to develop but offers long-term or lifelong protection.
- **Examples:** Recovery from diseases, vaccines.

Passive Immunity:

- Body receives ready-made antibodies.
- Offers immediate but temporary protection.
- **Examples:** Maternal antibodies, antiserum injections.

Summary:

Feature	Active Immunity	Passive Immunity
Antibody source	Produced by the individual's immune system	Given from outside (another individual or source)
Time to act	Slow	Fast (immediate)
Duration	Long-term	Short-term
Memory cells formed	Yes	No
Natural examples	Infection	Breast milk, placenta
Artificial examples	Vaccination	Antibody injection (e.g., antiserum)

2. Innate and Artificial Immunity

This section focuses on Active and Passive Immunity, both naturally and artificially acquired.

Active Immunity

• Definition:

Immunity that develops as a result of the body's own immune response to an antigen.

Types:

1. Natural Active Immunity:

Occurs after a natural infection (e.g., recovering from measles).

2. Artificial Active Immunity:

• Acquired through vaccination (e.g., polio vaccine).

Vaccines:

• A vaccine contains pathogens or their products to stimulate the immune system and induce immunity without causing disease.

Types of Vaccines:

1. Live Attenuated Vaccines:

• Weakened form of the live pathogen (e.g., MMR vaccine).

2. Inactivated Vaccines:

• Killed pathogens (e.g., hepatitis A vaccine).

3. Toxoid Vaccines:

• Inactivated toxins (e.g., tetanus vaccine).

4. Polysaccharide and Conjugate Vaccines:

• Target bacterial coating (e.g., pneumococcal vaccine).

Definition:
Immunity developed by receiving pre-formed antibodies or immune cells from another source (donor).

Types:

1. Natural Passive Immunity:
- Antibodies passed from mother to baby:
- Through placenta (during pregnancy)
- Through breast milk (after birth)

2. Artificial Passive Immunity:
- Injection of antibodies or immune cells from an immune donor (e.g., antiserum, monoclonal antibodies)

Comparison Table

Feature	Active Immunity	Passive Immunity
Response source	Body produces antibodies	Antibodies received from another individual
Time to develop	Slow (days to weeks)	Immediate
Duration	Long-term (can be lifelong)	Short-term (temporary)
Memorycells produced	Yes	No
Natural example	After infection	From mother (placenta/breast milk)
Artificial example	Vaccination	Antiserum or antibody injection

Cell-Mediated and Antibody-Mediated Immune Responses

Overview:
- The third line of defense is part of the adaptive immune system.
- It involves specific white blood cells known as lymphocytes.
- **Two types of lymphocytes:**
- T cells (Thymus-derived)
- B cells (Bone marrow-derived)

Development and Maturation:

Cell Type	Maturation Site	Discovery/Name Origin
B cells	Bone marrow	Named after Bursa of Fabricius in birds
T cells	Migrate to thymus to mature	"T" for Thymus, where they become functional

Types of Adaptive Immunity:

1. Cell-Mediated Immunity (CMI):

- Carried out by T lymphocytes.
- Especially effective against intracellular pathogens (e.g., viruses, some bacteria).
- Involves cytotoxic T cells (Tc) that kill infected cells.
- Also includes helper T cells (Th) which activate other immune cells.

2. Antibody-Mediated (Humoral) Immunity:

- Performed by B lymphocytes.
- Effective against extracellular pathogens (e.g., bacteria, toxins).
- B cells recognize antigens and produce antibodies.
- Antibodies bind specifically to antigens, marking them for destruction.

Summary Table:

Feature	Cell-Mediated Immunity (CMI)	Antibody-Mediated (Humoral) Immunity
Main Cells Involved	T cells (Tc, Th)	B cells
Target Pathogens	Intracellular (viruses, infected cells)	Extracellular (bacteria, toxins)
Key Mechanism	Direct cell killing, activation of immune cells	Antibody production
Maturation Site	Thymus	Bone marrow
Memory Cells Formed?	Yes	Yes

Cell-Mediated and Antibody-Mediated Immune Responses

1. Cell-Mediated Immunity (CMI) Response

- **Mediated by:** T-lymphocytes
- Do not produce or secrete antibodies.

Effective against:

- Viruses
- Intracellular parasites (e.g., some protozoa)
- Tumor cells
- Fungi

Mechanism:

- T cells identify infected or abnormal host cells and eliminate them.

Involves:

- **Cytotoxic T cells (Tc):** kill infected cells
- **Helper T cells (Th):** assist in activating Tc cells and other immune cells
- **Regulatory T cells (Treg):** suppress overactive immune responses
- **Memory T cells:** provide long-term immunity

2. Antibody-Mediated (Humoral) Immune Response

• **Mediated by: B-lymphocytes**
• Termed "humoral" because antibodies are secreted into body fluids (humours like blood and lymph).
• Effective against:
• Bacteria
• Toxins
• Free viruses (extracellular pathogens)

Mechanism:

• B cells recognize antigens via B cell receptor s (BCR).
• Once activated, they differentiate into plasma cells which produce antibodies.
• Some B cells become memory B cells for faster future responses.

Lymphocytes – Key Players

• Type of WBCs (20–40% of total WBCs).
• Originate from lymphoid stem cells in bone marrow.
• Found in blood and tissues.
• Two main types:
• T cells (develop in thymus)
• B cells (mature in bone marrow)

Characteristics:

Feature	T Cells	B Cells
Maturation Site	Thymus	Bone marrow
Surface Receptor	T Cell Receptor (TCR)	B CellReceptor (BCR)
Function	Cell-mediated immunity	Antibody-mediated (humoral) immunity
Response Type	Intracellular pathogens, cancer	Extracellular pathogens, toxins
Subtypes	Helper, Cytotoxic, Regulatory, Memory	Plasma cells, Memory B cells

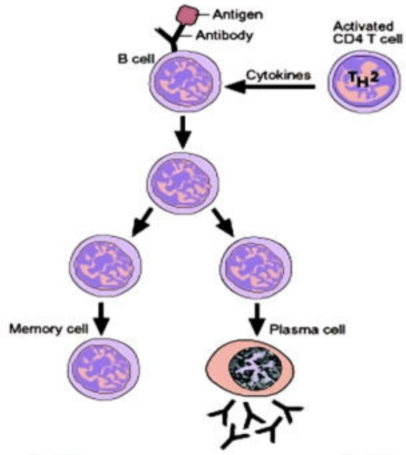
Lifespan:

• Most lymphocytes live a few weeks to months, but some live for years as memory cells, providing immunological memory.

Activation Signals for T Cells:

To become fully active, T cells require three signals:

1. TCR–MHC interaction (with antigen-presenting cells)
2. Costimulatory signals (e.g., CD28)
3. Cytokines (signaling molecules)



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Disorders of the Immune System

Allergies

• **Definition:** Allergies are exaggerated immune responses to harmless substances, called allergens (e.g., pollen, mold, pet dander, certain foods, drugs).

Mechanism:

• When allergens enter the body, **IgE antibodies** are produced by plasma cells.
• These antibodies bind to allergens and receptors on **mast cells** or **basophils**.
• Upon activation, these cells release **histamine**, causing inflammation.

Symptoms:

• Itching, rashes, runny nose, sore throat, sneezing, cough, etc.

Common Allergies:

• Rhinitis, asthma, eczema, dermatitis, hives, anaphylaxis, food allergies.

Treatment:

• Antihistamines and steroids are commonly used to manage symptoms.

Autoimmune Diseases

• **Definition:** Autoimmune diseases occur when the immune system mistakenly attacks its own tissues, failing to recognize them as "self".

Examples:

• Diabetes mellitus type-I, rheumatoid arthritis, psoriasis, etc.

Mechanism:

• Self-antibodies are produced that target and destroy the body's own tissues.

Possible Triggers:

• Bacterial/viral infections, drugs, toxins, pollutants, etc.

Common Affected Tissues:

• Joints, blood vessels, connective tissues, muscles.

Symptoms:

• Malaise (general ill feeling), low fever, fatigue, muscle aches, rashes.

Treatment:

• Antihistamine therapy can be used to control symptoms like rashes and runny nose.

Transplant Rejections

·• **Definition:** Transplant rejection occurs when the recipient's immune system attacks a transplanted organ due to genetic differences.

Mechanism:

• The **immune system** of the recipient recognizes the transplanted organ as foreign and attacks it.

Types of Immune Responses in Rejection:

• **Cell-mediated immunity (CMI)** and **humoral immunity** are involved in rejection.

Role of T-Cells in Transplant Rejection:

Mechanism:

• Donor tissue **dendritic cells** migrate to the recipient's lymphoid tissues and are recognized by the recipient's **TH cells** through antigen-presenting cells (APCs).

• Activated **TH cells** proliferate and activate **NK cells** and **Tc cells** to attack the grafted tissue.

Role of B-Cells in Transplant Rejection:

Mechanism:

• Activated **TH cells** signal **B cells** to differentiate into **plasma cells** that secrete antibodies.

• These antibodies attack the transplanted tissue, contributing to rejection.

Immunosuppressant Drugs for Transplant Recipients

Purpose:

• **Immunosuppressants** are given to transplant recipients to prevent organ rejection by suppressing the immune system.

Implications:

• Suppressing the immune system makes recipients more susceptible to infections.

• Side effects of immunosuppressants include:

• Fatigue, headache, nausea, vomiting, acne, diabetes, hypertension, hair loss, mouth sores, sepsis, osteoporosis, etc.